

Rationale and Methodology for Benchmarking V8 against V7

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Information Gaps & Caveats

Information Gaps: The prompt requests a robust benchmarking methodology but does not specify the computational constraints (e.g., time/memory limits for resampling large DTA networks) or the accepted statistical significance thresholds (α -levels) for the transportation policy domain.

Response Caveats: The recommended non-parametric tests (Bootstrapping, Wilcoxon) assume independent observations. Given the nature of transportation data, spatial and temporal autocorrelation will likely be present. In a true production environment, block-bootstrapping or spatial clustering corrections must be applied to prevent artificially narrow confidence intervals.

1. The Rationale for Benchmarking

When upgrading an econometric and simulation pipeline from static assumptions (V7) to empirical calibrations (V8), the benchmarking methodology must prove that the new architecture is **resilient** and significantly reduces prediction error against ground truth, rather than merely overfitting to the new survey microdata.

- **Model Parsimony & Justification:** Complex models are only preferred if they offer statistically significant improvements over simpler ones (Occam's Razor).
- **Spatial Convergence:** Ensuring that scaling up Census Block data to TAZ levels via Iterative Proportional Fitting (IPF) genuinely reduces spatial flow residuals compared to unconstrained V7 trip tables.
- **Out-of-Sample Generalization:** Verifying that V8's mode choice utility parameters generalize to real-world transit telemetry rather than memorizing the survey data.

2. Statistically Robust Benchmarking Methods

A. Bootstrapping for Non-Parametric Significance

Because transportation data (travel times, incident delays) is heavily skewed and rarely follows a normal distribution, standard parametric tests often fail.

- **Application:** Draw B bootstrap samples (e.g., $B = 1000$) of transit corridor segments. Calculate the prediction error for both V7 and V8 on each sample.
- **Significance Test:** Compute the difference in performance metrics (e.g., $RMSE_{V7} - RMSE_{V8}$) across the bootstrap samples to build a 95% confidence interval. If this interval is entirely above zero, the performance improvement in V8 is statistically significant.

B. Likelihood-Ratio (LR) Test (For Mode Choice)

Since V7's mode choice relies on restricted/static parameters and V8 introduces unconstrained parameters estimated via Maximum Likelihood Estimation (MLE), V7 is technically a "nested" model of V8.

- **Application:** Use the LR test to compare the goodness-of-fit of the nested logit models. Equation: $LR = -2(\mathcal{LL}_{V7} - \mathcal{LL}_{V8})$
- **Significance:** The LR statistic follows a Chi-square (χ^2) distribution. A p -value < 0.05 confirms that integrating microdata mathematically justifies the added model complexity.

C. Wilcoxon Signed-Rank Test (For Spatial Flows)

When comparing the TAZ-to-TAZ trip flows between the two models against the LEHD LODS ground truth, the error residuals will likely be paired but non-normally distributed.

- **Application:** Calculate the absolute error for every Origin-Destination pair in V7 and V8. The Wilcoxon test evaluates whether the median error in V8 is consistently and systematically lower than in V7, providing a rigorous alternative to a paired Student’s t-test.

3. Core Benchmarking Metrics

To comprehensively benchmark the models, metrics should be stratified across the distinct components of the pipeline:

Pipeline Component	V7 vs. V8 Benchmarking Metric	Target Benchmark
1. Mode Choice Utilities	McFadden’s Pseudo-R^2: Measures the improvement in log-likelihood over a null model.	V8 Pseudo- $R^2 > 0.25$ (indicating excellent empirical fit).
2. Trip & Spatial Flows	Kullback-Leibler (KL) Divergence: Measures how much the simulated probability distributions diverge from the LEHD LODS commuter flows.	$KL_{V8} < KL_{V7}$ (indicating lower entropy loss).
3. Travel Time / Speeds	Mean Absolute Error (MAE): The absolute minute-delay difference between simulated DTA skims and 511.org GTFS-RT ground truth.	V8 MAE reduction of at least 15% vs. V7.
4. Synthetic Demographics	Mean Absolute Percentage Error (MAPE): Evaluates how closely the expanded population matches Census marginals.	V8 MAPE $< 1.0\%$ across all Equity Priority Communities (EPCs).

4. Implementation Design (Hold-Out Validation)

To prevent data leakage, the benchmarking process should utilize a **Temporal Hold-Out** strategy. If the models are calibrated using microdata and flows from a specific temporal window (e.g., Q1 2026), the V7 vs. V8 benchmark tests must be executed against a completely unseen temporal snapshot (e.g., Q2 2026).